

UNIVERSITY OF ASIA PACIFIC
DEPARTMENT OF CSE

Course Code: CSE 402.

Course Title: Operating System Lab.

Experiment No: 7

Experiment Title: Implementation of CPU Disk Scheduling Algorithm

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Problem Statement:

Consider a disk scheduling system with an initial head position of 41 and the following pending disk requests: **[3, 60, 10, 25, 30, 4].**

Implement two disk scheduling algorithms, Shortest Seek Time First (SSTF) and First Come First Serve (FCFS), using Python. For both algorithms, determine the order in which requests are serviced and calculate the total disk head movement. Finally, analyze the obtained results to identify which algorithm provides better performance in terms of reducing disk head movement.

Source Code:

```
req = [3, 60, 10, 25, 30, 4]
head = 41
n = len(req)

done = [0] * n
current = head
total_movement = 0
seq = [current]

for i in range(n):
    x = -1

    min_distance = 9999


    for j in range(n):
        if done[j] == 0:
            dist = abs(req[j] - current)

            if dist < min_distance:
                min_distance = dist
                x = j
```

```
    movement = min_distance

    print(f"{current} -> {req[x]} = {movement}")


    total_movement += movement
    current = req[x]
    seq.append(current)
    done[x] = 1


print("Total Movement: ", total_movement)
print("Sequence: ", seq)



print()
requests = [3, 60, 10, 25, 30, 4]
head = 41

total_movement = 0
current_head = head

print("FCFS Disk Scheduling")
print("Initial Head:", head)
print("Request Sequence:", requests)
print()

for request in requests:
    movement = abs(request - current_head)
    print(f"{current_head} -> {request} = {movement}")

    total_movement += movement
    current_head = request
print()
print("Total Seek Operation / Total Head Movement =", total_movement)
```

Output:

```
PS D:\New folder> & C:\Users\lab\AppData\Local\Programs\Python\Python314
41 -> 30 = 11
30 -> 25 = 5
25 -> 10 = 15
10 -> 4 = 6
4 -> 3 = 1
3 -> 60 = 57
Total Movement:  95
Sequence:  [41, 30, 25, 10, 4, 3, 60]

FCFS Disk Scheduling
Initial Head: 41
Request Sequence: [3, 60, 10, 25, 30, 4]

41 -> 3 = 38
3 -> 60 = 57
60 -> 10 = 50
10 -> 25 = 15
25 -> 30 = 5
30 -> 4 = 26

Total Seek Operation / Total Head Movement = 191
PS D:\New folder>
```

Discussion:

The experiment evaluates how request selection affects the distance travelled by a disk head. Both algorithms use the same initial position and request queue, but their servicing strategies produce different results.

In FCFS, the disk head follows the requests in their original order, resulting in a total movement of 191 tracks. SSTF instead prioritizes the nearest available request and achieves a total movement of only 95 tracks. Consequently, SSTF saves 96 tracks of movement, representing approximately 50.2618% less movement than FCFS for this dataset.

These findings demonstrate that choosing requests based on their distance from the current head position can significantly reduce overall movement. However, SSTF does not guarantee that every request will receive service quickly, particularly when some requests remain far away from the head.

Conclusion:

The experiment successfully implemented and evaluated SSTF and FCFS disk scheduling algorithms using Python. The calculated total head movement was 95 tracks for SSTF and 191 tracks for FCFS, demonstrating a clear difference in their performance for the given request queue. SSTF is the more efficient choice in this case because it reduces head movement by 96 tracks compared with FCFS. Nevertheless, FCFS offers a straightforward and predictable servicing order, while SSTF focuses on minimizing the distance travelled. Therefore, SSTF is preferable when reducing total head movement is the primary objective, although fairness and the possibility of request starvation should also be considered when selecting a disk scheduling algorithm.

GitHub:

https://github.com/boishakhi73/CSE-402-Operating-System-Lab